

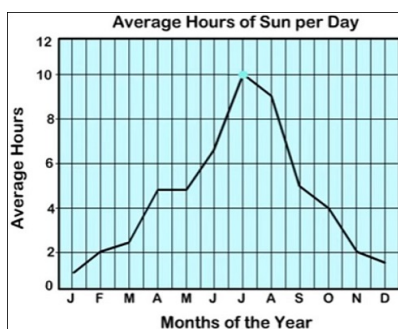
Q.1 Another name of mean in statistics is

- A. Mode
- B. None of these
- C. Median
- D. Expected value**

Q.2 Contribution Margin = Sale - Variable Cost

- A. False
- B. True**

Q.3 The graph below shows the average hours of sun per day over a period of one year. Which other month had the same number of average hours of sun per day as February?



- A. November**
- B. July
- C. January
- D. December

Q.4 A trimmed mean is also known as:

- A. Weighted Mean
- B. Truncated Mean**
- C. Median Mean
- D. Average Mean

Q.5 In a Pie chart, usually, the arrangement of the sectors is

- A. None of these
- B. Anti-clockwise**
- C. Clockwise
- D. Pairwise

Q.6 Degrees are used to present percentages in _____

- A. none of these
- B. bar graph
- C. Sector graph**
- D. Line graph

Q.7 In Positively Skewed Distribution, which of the following is true

- A. Mean > mode > median
 - B. Mode > median > mean
 - C. None of these
 - D. Mean > median > mode**
-

Q.8 If there are three intervals in the bins_array, how many cells should you select to display the FREQUENCY results?

- A. 5
 - B. 4
 - C. 2
 - D. 3**
-

Q.9 Histogram is also an example of graph.

- A. True**
 - B. false
-

Q.10 Break even point (BEP) in units is calculated as

- A. Contribution Margin (CM)/Fixed cost (FC)
 - B. Contribution Margin (CM)+Fixed cost (FC)
 - C. Contribution Margin (CM)-Fixed cost (FC)
 - D. Fixed cost (FC)/ Contribution Margin (CM)**
-

Q.1 Last year a small statistical consulting company paid each of its five statistical analysts \$50,000 each, and the senior statistician/owner \$270,000. The number of employees the mean salary is:

- B. 5
- C. 4
- D. 7**

Q.2 Which of the following is true?

- A. None of these.
- B. Arithmetic Mean is based on all and Mode is on the middle values.
- C. Arithmetic Mean is based on all and median on the middle value.**
- D. Median based on most frequent value and mode on middle value.

Q.3is used in compounding such as investments that are accumulated over a period of time

- A. Mid range
- B. Harmonic mean
- C. Geometric mean**
- D. Arithmetic mean

Q.4 In a set of 10 observations the mean is 20 and the median is 15. There are 2 values that are 6, and all other values are different. What is the mode?

- A. 15
- B. 20
- C. 16
- D. 6**

Q.5 Mode of the data 25,35,21,56,25,56,65 is

- A. 21
- B. 35
- C. 25 & 56**
- D. 65

Q.6 Which of the following is the measure of central tendency?

- A. Variance analysis
- B. Arithmetic Mean**
- C. None of these
- D. Standard deviation

Q.7 A type of graph that shows change over time is a

- A. double-bar graph
 - B. line graph.**
 - C. circle graph
 - D. bar graph.
-

Q.8 The formula for class width =

- A. Range / no of classes**
 - B. Mode / no of classes
 - C. Mean / no of classes
 - D. Median /no of classes
-

Q.9 Under which of the following conditions for three terms a, b, c , their all central tendencies (mean, median & mode) are equal

- A. $a > b > c$
 - B. $a = b = c$**
 - C. $a > b$ but $b < c$
 - D. $a < b < c$
-

Q.10 The mode of the data 2.7, 3.5, 4.9, 5.1, 8.3 is

- A. There is no mode for this data**
 - B. 3.5
 - C. 5.1
 - D. 4.9
-

Q.1 The average of the first and third quartiles is called...

- A. Interquartile range
 - B. Range
 - C. Midhinge**
 - D. Midrange
-

Q.2 If BEP in rupees is Rs. 25400, the selling price S is RS. 200 per unit and CM is 30, then what is FC per period?

- A. Rs. 3830
 - B. Rs. 3820
 - C. Rs. 3840**
 - D. Rs. 3810
-

Q.3 If the break even point in units is 220 and the production capacity is 660, then break even point as a percentage of the capacity is

- A. 33.34%**
 - B. 300%
 - C. 50%
 - D. 3%
-

Q.4 If fixed cost is 8550 and contribution margin is 25 then break even point in unit will be

- A. 342**
 - B. 8525
 - C. 213750
 - D. 8575
-

Q.5 If FC per period is 3500, S is RS. 190 per unit and CM is 70, then what is the BEP in rupees?

- A. 7500
 - B. 6500
 - C. 9500**
 - D. 8500
-

Q.6 Which of the following is NOT a requirement for class intervals?

- A. Classes must be mutually exclusive
 - B. Classes must have gaps between intervals**
 - C. Classes must be equal in width
 - D. Classes must be exhaustive
-

Q.7 If the sales-agent sells 200 bulbs for Rs 24 each, what is the net sale?

- A. 4800**
 - B. 0.12
 - C. 8.33
 - D. None of these
-

Q.8 BEP in units = FC/CM

- A. True**
 - B. False
-

Q.9 Calculate the value of BEP given that $FC = \text{Rs. } 3136$ and $CM = \text{Rs. } 20$

- A. None of these
 - B. 0.0064
 - C. 62720
 - D. 156.80**
-

Q.10 Another name of mean in statistics is

- A. Expected value**
 - B. None of these
 - C. Median
 - D. Mode
-

Q.1 Which of the following is true?

- A. $M > G.M > H.M$
- B. $M < G.M < H.M$**
- C. None of these.
- D. $M = G.M = H.M$

Q.2 A type of graph that shows change over time is a

- A. line graph.**
- B. double-bar graph
- C. bar graph.
- D. circle graph

Q.3 If FC is Rs 4430 and the BEP in units is 89, then Contribution Margin is

- A. Rs 45.6
- B. Rs 50.8
- C. Rs 49.8**
- D. Rs 40.5

Q.4 A picture graph uses to show data.

- A. Pictures**
- B. Lines
- C. Points
- D. Sectors

Q.5 The statement

"~~Mean~~**Does not get affected by the extreme values or outliers in the data**"
is

- B. False**

Q.6 The measure of central tendency which divides the data into two equal parts is called.....

- A. Mode
- B. Mean
- C. Geometric mean
- D. Median**

Q.7 Class width of the given data is defined as ratio of

- A. Range and mid point/value
- B. Range and number of values in the given data
- C. None of these
- D. Range and the number of classes**

Q.8 When we add frequency of the second interval to the frequency of the first interval, then the for the second interval is obtained.

- A. Relative frequency
 - B. None of these
 - C. Cumulative Frequency**
 - D. Percent Relative Frequency
-

Q.9 A full circle hasdegrees.

- A. 360**
 - B. 460
 - C. 260
 - D. 560
-

Q.10 In the AVERAGEA function, a logical value of TRUE is treated as _____.

- A. 1**
 - B. n
 - C. 0
 - D. Ignored
-

Q.1 In MS Excel, which of the following is ignored by the MEDIAN function?

- A. All of these**
 - B. Cells containing numbers
 - C. Cells containing text
 - D. Cells with the value zero
-

Q.2 A full circle hasdegrees.

- A. 360**
 - B. 260
 - C. 460
 - D. 560
-

Q.3 How is the number 26 represented in a stem-and-leaf plot?

- A. Stem: 20, Leaf: 6
 - B. Stem: 26, Leaf: 0
 - C. Stem: 2, Leaf: 6**
 - D. Stem: 6, Leaf: 2
-

Q.4 At the break even point

- A. cash inflow is not equal to cash outflow
 - B. cash inflow is greater than cash outflow
 - C. cash inflow is less than cash outflow
 - D. cash inflow is equal to cash outflow**
-

Q.5 Which of the following is NOT a typical step in tabulating numerical data?

- A. Calculating the mean of the data.**
 - B. Determining the number of classes.
 - C. Sorting the data in ascending order.
 - D. Defining class boundaries.
-

Q.6 When we calculate the mode of the data then it is necessary that first we

- A. d) a and b both are wrong
 - B. c) a and b both are correct
 - C. b) Arrange the data in descending order
 - D. a) Arrange the data in ascending order**
-

Q.7 In Positively Skewed Distribution, which of the following is true

- A. None of these
 - B. Mode > median > mean
 - C. Mean > mode > median
 - D. Mean > median > mode**
-

Q.8 For some data you are given Maximum value = 96 , Minimum Value = 23 , Range = 73, number of classes selected between 5 and 15 Then class width will be

- A. 8
 - B. 5
 - C. 13
 - D. All options 1,2,3 are possible.**
-

Q.9 A pictograph also called

- A. Column Graphs
 - B. Conversion Graphs
 - C. Pictogram**
 - D. Histograms
-

Q.10 The Harmonic Mean of the data 8, 5, 3 is

- A. None of these
 - B. 5.44
 - C. 3.44
 - D. 4.55**
-

Q.1 A is a graphical display of a frequency table.

- A. Sector
 - B. histogram**
 - C. Ogive
 - D. frequency polygon
-

Q.2 How is the median calculated if a dataset has an even number of values?

- A. It cannot be determined.
 - B. It is the average of the two middle values.**
 - C. It is the middle value.
 - D. It is the most frequent value.
-

Q.3 What are the central tendencies?

- A. Mean, median, mode and range
 - B. Mean, median, mode and midrange**
 - C. none of them
 - D. Mean, median, mode and harmonic mean
-

Q.4 For some data you are given Maximum value = 96 , Minimum Value = 23 , Range = 73, number of classes selected between 5 and 15 Then class width will be

- A. All options 1,2 ,3 are possible.**
 - B. 13
 - C. 8
 - D. 5
-

Q.5 The measure of central tendency which divides the data into two equal parts is called.....

- A. Geometric mean
 - B. Mean
 - C. Median**
 - D. Mode
-

Q.6 In a stem-and-leaf display of a decimal number, what does the "leaf" represent?

- A. The tens place
 - B. The units place**
 - C. The middle digit
 - D. The sum of all digits
-

Q.7 What type of formula is the FREQUENCY function?

- A. Array formula**
 - B. Lookup formula
 - C. Conditional formula
 - D. Logical formula
-

Q.8 Frequency polygon is a obtained from a frequency distribution.

- A. A line graph**
 - B. None of these
 - C. Pareto Chart
 - D. Bar Chart
-

Q.9 How does the FREQUENCY function handle text in the data_array?

- A. Treats it as zero
 - B. Returns an error
 - C. Converts it to numeric values
 - D. Ignores it**
-

Q.10 In a dataset, which of the following would impact the mean most?

- A. Extreme low values
 - B. Extreme values**
 - C. Negative values
 - D. Extreme high values
-

Q.1 use the division of a circle into different sectors.

- A. Sector
 - B. Line
 - C. column and Bar
 - D. picture
-

Q.2 A line graph shows plotted on a graph.

- A. Lines
 - B. Sectors
 - C. Picturs
 - D. Points
-

Q.3 A is a graphical display of a frequency table.

- A. frequency polygon
 - B. Ogive
 - C. Sector
 - D. histogram
-

Q.4 A bar graph usesto show data.

- A. Points
 - B. Bars
 - C. Lines
 - D. Pictures
-

Q.5 The statement

"Median does not get affected by extreme values or outliers in the data" is ...

- B. True
-

Q.6 A full..... has 360 degrees.

- A. Circle
 - B. Sphere
 - C. Cone
 - D. Parabola
-

Q.7 Ogives are used to determine at different values.

- A. Frequency
 - B. The cumulative Frequency
 - C. None of these
 - D. Frequency distribution
-

Q.8 Mode of a set of values can be

- A. Two
 - B. Three
 - C. Exactly one
 - D. More than one**
-

Q.9 How does the FREQUENCY function handle text in the data_array?

- A. Converts it to numeric values
 - B. Treats it as zero
 - C. Returns an error
 - D. Ignores it**
-

Q.10 The most frequent value in the data is called

- A. Mode**
 - B. Mean
 - C. Midrange
 - D. Median
-

Q.1 Text method presents _____

- A. Data in rows
 - B. Charts and graphs
 - C. tables with rows and columns**
 - D. none of these
-

Q.2 If the Contribution Margin (CM) is 30 and the Selling Price S is Rs. 210 per unit, then what is the Variable Cost (VC)?

- A. Rs. 170
 - B. Rs. 160
 - C. Rs. 150
 - D. Rs. 180**
-

Q.3 In a histogram, what do the widths of the bars represent?

- A. Cumulative frequencies
 - B. Midpoints of the classes
 - C. Class frequencies
 - D. Class intervals**
-

Q.4 Frequency Distributions involve

- A. All of these**
 - B. Polygons
 - C. Histograms
 - D. Tables
-

Q.5 In the AVERAGE function, which of the following is included in the calculation?

- A. Text in the cells
 - B. Cells with the value 0**
 - C. Logical values
 - D. Empty cells
-

Q.6 The median of the numbers 2, 4, 7, 12 is

- A. 4
 - B. 7
 - C. There is no median for this data
 - D. 5.5**
-

Q.7 If the mean of the data is 50 and sum of the data is 450 then the number of observations in the data will be

- A. 500
 - B. None of these
 - C. 22500
 - D. 9**
-

Q.8 Define Break Even Point.

- A. It is a point at which neither a profit nor loss is made.**
 - B. It is a point at which either a profit or loss is made.
-

Q.9 If Variable Cost is Rs.120 and contribution margin is Rs 30, then sale will be -----

- A. 140
 - B. 130
 - C. 150**
 - D. 120
-

Q.10 The most frequent value in the data is called

- A. Mode**
 - B. Median
 - C. Mean
 - D. Midrange
-

Q.1 The moving averages of the Prices 55,60,65,70 are

- A. 65,65
 - B. 70,60
 - C. 60, 65**
 - D. 70, 75
-

Q.2 The correlation coefficient between two variables, r:

- A. is expected to be zero when there is no relationship between the variables.
 - B. must lie between -1 and +1 inclusive.**
 - C. can only have a valid significance test carried out when one variable follows a Normal Distribution
 - D. depends on which of the two variables is chosen to be the dependent variable.
-

Q.3 The sign of regression coefficient is associated with

- A. co-variance of x and y**
 - B. variance of x
 - C. variance of y
 - D. none of these
-

Q.4 If the value of r is 0.8 ,then the coefficient of determination is

- A. 80%
 - B. 67%
 - C. 75%
 - D. 64%**
-

Q.5 Removing the highest & lowest actual-to-moving average values when computing seasonal index for annual data reduces.

- A. Extreme cyclical variations
 - B. Secular trend
 - C. All of these
 - D. Seasonal variations**
-

Q.6 Averages that are calculated in method of moving average is called _____.

- A. Arithmetic mean**
 - B. Mode
 - C. Geometric mean
 - D. Median
-

Q.7 The correlation co-efficient varies from

- A. 0 to 1
 - B. -1 to 1**
 - C. None of these
 - D. -1 to 0
-

Q.8 Variance or standard deviation can be zero only if

- A. all the values in the data are equal**
 - B. all the values in the data form in descending order
 - C. all the values in the data are in ascending order
 - D. all the values in the data are unequal
-

Q.9 The main points of interest for simple linear regression are:

- A. all given options.**
 - B. Coefficient of determination.
 - C. STEM-Standard Error of mean and T-Statistic.
 - D. Correlation Coefficient.
-

Q.10 Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

- A. Decomposition model
 - B. Trend analysis
 - C. Exponential smoothing method**
 - D. Multiplicative model
-

Q.1 A relationship between variables that can be represented by a straight line equation is

- A. Non linear equation
 - B. Population regression model
 - C. Simple linear regression model**
 - D. Non linear regression model
-

Q.2 We use to represent relationship between two variables by fitting a linear equation to observed data.

- A. Covariance
 - B. Multiple regression
 - C. Correlation
 - D. Linear regression**
-

Q.3 Value of Correlation coefficient always

- A. lesser than -1
 - B. between 0 & 1
 - C. greater than +1
 - D. between +1 & -1**
-

Q.4 What does Multiple R represent in regression analysis?

- A. Square of the R-squared value.
 - B. Random error of the model.
 - C. Proportion of variance explained by the model.
 - D. Correlation coefficient between the predicted and observed values.**
-

Q.5 Which of the following is used most frequently measures of dispersion because of its statistical properties and interpretive value?

- A. Mean
 - B. Mode
 - C. Range
 - D. Standard deviation**
-

Q.6 Seasonal variation is defined as ____.

- A. Actual + trend
 - B. Actual $\times$ trend
 - C. None of these
 - D. Actual - trend**
-

Q.7 There are two graphs A1 and A2. The graph A1 has $r = 0.9$ and the graph A2 has $r = 0.5$, then what do you conclude?

A. A1 is less scattered than A2.

B. A1 and A2 are equally scattered.

C. A1 is more scattered than A2.

D. None of these

Q.8 The daily sale of an item in the market.

Averages

2 Morning 80 75.7

3 Daytime 60 70.3

drawn series of sales 82 82.3

Afternoon 70 75.7

Number of sales is increasing

Evening 80 68.3

B. No result can be drawn.

Day

By looking at the table, what do you

D. The sale is decreasing.

Q.9 The intercept of a line passing through origin is

A. zero

B. does not exists

C. finite

D. positive

Q.10 Mean deviation of the set 2,4,6 is

A. 2.8

B. 1.33

C. 01

D. 4

Q.1 If $S_{xy} = -5.5$, $S_y = 2.35$ and $r = -0.08$, then Standard deviation of X is

- A. 29.25
- B. 15.77
- C. 12.85
- D. 20.47

Q.2 The sum of deviation of the observed values of Y_i from the regression line is always equal to

- A. None
- B. 0
- C. -1
- D. 1

Q.3 The standard error of estimate is

- A. in squared units of the independent variable.
- B. based on squared deviations from the regression line.
- C. may assume negative values.
- D. All the given choices are correct.

Q.4 The minimum number of points required to calculate the intercept of a straight line is/are

- A. one or three
- B. three
- C. two
- D. one

Q.5 The mode of the words in the word CORRELATION is

- A. O
- B. R
- C. Both R and O
- D. L

Q.6 In method of least square we

- A. maximize the sum squares of residuals
- B. maximize the sum of residuals
- C. minimize the sum of residuals
- D. minimize the sum squares of residuals

Q.7 For a data, mean < median then its distribution is

- A. asymmetrical
 - B. right asymmetrical
 - C. left asymmetrical
 - D. symmetrical
-

Q.8 If $a = 5$, $b = 1$, then an estimating regression line is

- A. $Y = 5 + X$.
 - B. $Y = 1 + 5X$.
 - C. $X = 5 - Y$.
 - D. $X = 5 + 5Y$.
-

Q.9 If $\text{Summation}(X) = 55$, $\text{Summation}(Y) = 34$, $\text{Summation}(XY) = 223$, $\text{Summation}((X^2)) = 525$, $n = 7$
Slope of data is
Then

- A. -0.47538
 - B. 0.6978
 - C. -0.8910
 - D. 0.3187
-

Q.10 The square of the correlation coefficient is called the

- A. coefficient of determination
 - B. covariance
 - C. big R
 - D. variance
-

Q.1 The point where a straight line cuts the x-axis is called

- A. starting point
- B. slope
- C. y-intercept
- D. x-intercept**

Q.2 Regression line always passes through

- A. Mean of data**
- B. Median of data
- C. Both i. & ii.
- D. None

Q.3 A simple linear regression model is an equation that describes the straight-line relationship between a dependent variable and an independent variable.

- A. False
- B. True**

Q.4 If for a regression line of a given data set, intercept = 5, $\bar{Y} = 10$, $\bar{X} = 5$ then the value of slope will be equal to

- A. 5/9
- B. 1**
- C. 5
- D. 9/5

Q.5 Which inputs are needed for the RSQ function in Excel?

- A. Two arrays of data (independent and dependent variables).**
- B. One array of data.
- C. Only the independent variable.
- D. Only the dependent variable.

Q.6 A relationship between the three variables X, Y AND Z is described as $Z = a + bX + cY$ is called

- A. Quadratic Equation
- B. Linear Equation**
- C. Linear regression
- D. Non Linear Equation

Q.7 If the dependent variable increases with the independent variable then the coefficient of correlation is

- A. -0.5 to 0
- B. 0 to -2
- C. None of these
- D. 0 to 1**

Q.8 If the standard deviation of a population is 9, the population variance is:

- A. 21
 - B. 81**
 - C. 9
 - D. 3
-

Q.9 The square of the correlation coefficient is called the

- A. covariance
 - B. big R
 - C. coefficient of determination**
 - D. variance
-

Q.10 In the scatter diagram, clustering of points around a straight line indicates

- A. none of these
 - B. non-linear regression
 - C. curvilinear regression
 - D. linear regression**
-

Q.1 In the equation $Y=2+3X$ the intercept is

- A. 2
 - B. 4
 - C. 3
 - D. 1
-

Q.2 If the value of r is 0.79 then there exist correlation

- A. None of these
 - B. strong negative
 - C. strong positive linear
 - D. weak positive
-

Q.3 The line with slope zero is

- A. horizontal line
 - B. parallel
 - C. oblique
 - D. vertical line
-

Q.4 If the equation of regression line is $y = 6$, then which of the following will be true for it.

- A. The line is parallel to y-axis
 - B. The line passes through the origin
 - C. Slope of the line is equal to 6
 - D. The line is parallel to x-axis
-

Q.5 Scatter plots are used only for quantitative variables.

- A. True
 - B. False
-

Q.6 If the standard deviation of a population is 9, the population variance is

- A. 81
 - B. 21.35
 - C. 3
 - D. 9
-

Q.7 If $y=2+0.6x$ then value of y-intercept is

- A. 2
 - B. 0
 - C. 1.2
 - D. 0.6
-

Q.8 For the least squares regression line $y = 0.9 + 1.7x$, ___ and ___ are slope and intercept respectively.

- A. 0.9 and 1.7
 - B. 2.6 and 0
 - C. 0 and 1
 - D. 1.7 and 0.9**
-

Q.9 What does the slope of the linear regression line represent?

- A. All of these.
 - B. Ratio of the change in the dependent variable to the change in the independent variable.**
 - C. Vertical distance divided by the horizontal distance between any two points on the line.
 - D. Rate of change along the regression line.
-

Q.10 Which method helps to draw a line between the set of scattered points

- A. Correlation method
 - B. Least square method**
 - C. Least fit method
 - D. Regression method
-

Q.1 For small to moderate sample size the sample standard deviation is _____.

- A. Unbiased
- B. None of these.
- C. Underestimated**
- D. Overestimated

Q.2 Seasonal variation repeat themselves regularly over short term greater than a year.

- A. False**
- B. True

Q.3 If the correlation coefficient is 0, what does it imply about the relationship between two variables?

- A. No relationship**
- B. Perfect positive relationship
- C. Perfect negative relationship
- D. Non-linear relationship

Q.4 Which of the following is not based on squared deviations from the regression line?

- A. The coefficient of determination.
- B. The standard error of estimate.
- C. The standard deviation.**
- D. The coefficient of correlation.

Q.5 Suppose the slope of regression line is -20 and the intercept is 9, then the equation of regression line will be -----

- A. $y - 20x + 9 = 0$
- B. $y - 20x - 9 = 0$
- C. $y + 20x - 9 = 0$**
- D. $y + 20x + 9 = 0$

Q.6 Variance or standard deviation can be zero only if

- A. all the values in the data are in ascending order
- B. all the values in the data are unequal
- C. all the values in the data are equal**
- D. all the values in the data form in descending order

Q.7 Square of the deviation will be greater if

- A. value is far from mean**
- B. value is close to mean
- C. value is coincident with mean
- D. value lies on the last quartile of data

Q.8 Choose the correlation that describes the strongest relationship between two variables

- A. - .25
 - B. .25
 - C. - .70**
 - D. .60
-

Q.9 If all of the y-values are the same, what will the SLOPE function return?

- A. #DIV/0! error
 - B. Undefined
 - C. 0**
 - D. 1
-

Q.10 Analysis of Variance (ANOVA) is a test for equality of:

- A. variances
 - B. proportions
 - C. only two parameters
 - D. means**
-

Q.1 Scatterplots, like histograms, are a good visual means to understanding patterns of

A. bivariate numerical data

B. Trivariate numerical data

C. univariate numerical data

D. None of these

Q.2 If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line

A. slope, intercept

B. slope, regression coefficient

C. intercept, slope

D. radius, intercept

Q.3 If the estimating equation is $Y = a - bX$, which of the following is true

A. The y intercept is b

B. There is inverse relationship

C. Slope of line is positive

D. all of these

Q.4 My estimated regression line is $Y = 17 + 4X$. The intercept is equal to:

A. 4

B. 17

C. 13

D. 21

Q.5 Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

A. If $X = 5$, then $Y = 15$.

B. The equation crosses the Y-axis at 3.

C. X and Y are inversely proportional

D. The dependent variable increases 2.1 for an increase of 1 in X

Q.6 As the regression line equation is $y = ax + b$, where a is

A. The slope

B. The x-intercept

C. The y-intercept

D. $y = x$

Q.7 The regression model explains the _____ between two or more variables in the given population.

- A. Connection
 - B. Association
 - C. Relation
 - D. All of these**
-

Q.8 The moving averages can be used for -----.

- A. Middle value.
 - B. mode
 - C. forecasting purposes**
 - D. Median
-

Q.9 The equation for the intercept of the regression line is $Y=a+bX$ where b is

- A. Equation
 - B. Dependant variable
 - C. Slope**
 - D. Intercept
-

Q.10 In 10 matches, Player A has scored a total of 425 runs with a variance of 36, whereas Player B, a football player, has scored a total of 18 goals with a variance of 2.25. Which player exhibits more consistency?

- A. Both are consistent.
 - B. Player B
 - C. Can't conclude
 - D. Player A**
-

Q.1 The mathematical equation describing the liner relation is

- A. $Y = C$
 - B. $Y = x$
 - C. $Y = a + bx^2 + C$
 - D. $Y = a + bx$**
-

Q.2 The slope of the regression line may be

- A. Positive
 - B. Negative
 - C. All of these**
 - D. Zero
-

Q.3 A simple linear regression equation.

- A. Is not affected by changes in the units in which variables are measured.
 - B. Always describes a line with zero slope.
 - C. Describes a line which goes through the point defined by the means of the two variables.**
 - D. Always describes a line which goes through the origin.
-

Q.4 In scatter diagram the dependent variable is plotted on the

- A. x-axis
 - B. y-axis**
-

Q.5 As the regression line equation is $y = ax + b$, where a is

- A. The y-intercept
 - B. The slope**
 - C. The x-intercept
 - D. $y = x$
-

Q.6 If the standard deviation is small, then many data points will be -----.

- A. Vanished.
 - B. Far away from the mean
 - C. On the mean point
 - D. Very close to the mean**
-

Q.7 If the INTERCEPT function returns a value of 0, what does it mean?

- A. Dependent and independent variables are uncorrelated.
 - B. Regression line passes through the origin.**
 - C. Slope of the line is zero.
 - D. No relationship between the variables.
-

Q.8 The regression model explains the _____ between two or more variables in the given population.

- A. Connection
 - B. Relation
 - C. Association
 - D. All of these**
-

Q.9 Mean deviation of the set 2,4,6 is

- A. 01
 - B. 1.33**
 - C. 2.8
 - D. 4
-

Q.10 The intercept of a line passing through origin is

- A. finite
 - B. does not exists
 - C. positive
 - D. zero**
-

Q.1 Rule to draw a scatter diagram is

- A. None
- B. Regression variable along X-axis and dependent variable along Y-axis
- C. Dependent variable along Y-axis and Regression variable along X-axis**
- D. Both i. & ii.

Q.2 The value of coefficient of variation can be

- A. Greater than 100
- B. Positive
- C. All of the these**
- D. Negative

Q.3 Which of the following is used most frequently measures of dispersion because of its statistical properties and interpretive value?

- A. Mean
- B. Mode
- C. Standard deviation**
- D. Range

Q.4 The equation of regression line is $y + 5x = 0$. What will be the intercept?

- A. 0**
- B. -5
- C. 5
- D. 1

Q.5 If there is a indirect relationship between rainfall & yield of crops then

- A. Yield is equal to rainfall
- B. Yield is lower if rainfall is less
- C. Yield is higher if rainfall is higher**
- D. Yield is higher if rainfall is less

Q.6 The following results were obtained from a simple regression analysis:

$$r^2 = .6774 \quad s = .2934$$

(dependent variable = 2024 equal to

- A. 0.6774
- B. 37.2895
- C. 0.2934
- D. -1.2024**

For each unit change in X (independent variable), the estimated change in Y

Q.7 Seasonal variations are _____ variation.

- A. Sudden
 - B. Short term**
 - C. None of theses.
 - D. Long term
-

Q.8 Plot scatter diagram helps us in finding the relationship b/w

- A. Median & Mode
 - B. Constant & a variable
 - C. Three variables
 - D. Two variables**
-

Q.9 If the dependent variable increases with the independent variable then the coefficient of correlation is

- A. 0 to -1
 - B. 0 to - 0.5
 - C. 0 to 1**
 - D. 0 to -2
-

Q.10 Which of the following is a measure of central tendency?

- A. Mode**
 - B. Mean deviation
 - C. Range
 - D. Standard deviation
-